

Muhammad Hassan

 [mhassandata](#) |  [dshassan](#) |  [Gmail](#) |  [Cell number](#)

SUMMARY

Muhammad Hassan is a self-taught Data Scientist with a strong background in leveraging data for business solutions, specializing in analytics, visualization, and predictive modelling. His expertise is widely recognized, and he holds the title of Kaggle Notebook Expert, actively contributing insights to the data science community. Muhammad is particularly interested in exploring the evolving field of generative AI and Large Language Models (LLMs) to enhance his professional knowledge and skills

WORK EXPERIENCE

Data Scientist Engineer at JazzCash Pakistan

JAN 2026 - till

The following major tasks are performed here.

Core Responsibilities.

Scalable Data Engineering: Design and maintain high-performance **ETL/ELT pipelines** using Python and Talend, implement advanced **SQL analytics**, and apply optimized **data warehouse modeling** techniques.

AI and Machine Learning Enablement: Build and manage data pipelines for **AI/ML workflows**, including feature engineering, model support, and intelligent data validation for analytics and predictive systems.

Automation and Monitoring: Automate routine data operations, job orchestration, and quality monitoring using **Linux Bash scripting** and intelligent automation to ensure reliability and scalability.

AI Engineer at NexGenAI Security System

JAN 2025 - Dec 2025

The following major tasks are performed here.

Core Responsibilities.

AI-powered access control: Upgrade the legacy CCTV Cameras with facial recognition.

Live Video and Breach Alerts: Provides live camera feeds enhanced with smart overlays and automated breach detection, enabling immediate response. **Comprehensive Security Tools and**

Geo-Fencing : Enhances existing CCTV with panic alarms, geo-fencing for automatic system arming/disarming, and remote monitoring via mobile and desktop apps

Professional Development and Training: Meta AI Master Trainer

Sep 2025 - Till

Master Trainer, AI Soft Skills.

Meta AI Upskilling for Pakistan (HEC and atomcamp) — Islamabad — Sep 2025

Train the Non-Tech Faculty of Pakistani University:

- Selected as one of 30 AI experts for the national 'Meta AI Upskilling for Pakistan' Master Trainer program, supported by HEC and Meta.
- Trained in advanced prompt engineering, responsible AI, and curriculum design for training non-technical faculty.
- Co-developed assessment frameworks and demonstrated teaching proficiency by delivering a model lesson to peers.

Machine Learning Engineer at Clinical Pearl

July 2024 - Jan 2025

The following major tasks are performed here.

Fine-tuning of randomized clinical trials using various language models including OpenAI, LLAMA, and Mistral.

Implementing RAG (Retrieval-Augmented Generation) system for advanced medical chat functionalities focused on diseases and pharmaceuticals utilizing specialized medical language models.

Developing multi-modal RAG systems integrating diverse data types and multi-modal language models.

Utilizing generative AI techniques to innovate in healthcare, enhancing diagnostic and treatment processes

AI Engineer at Smart Forum

Dec 2023 - Aug 2024

As a former AI Engineer at Smart Forum, I led the development of cutting-edge Generative AI applications across the Health, Education, and Media sectors. I spearheaded a team dedicated to crafting innovative solutions that disrupted traditional industry workflows. Beyond technical execution, I focused on the strategic application of technology to drive national economic growth and foster industrial advancement.

Machine Learning Trainer at Ai-Lounge

May 2023 - Nov 2023

Led the training of over 350 students and professionals from the northern region of Pakistan in emerging technologies such as Artificial Intelligence (AI), Data Science, and Blockchain. This initiative, in collaboration with the IT Department of the Government of Gilgit-Baltistan, focused on empowering individuals with cutting-edge skills and was conducted under the auspices of NUST. Played a pivotal role in developing curricula, mentoring participants, and driving the adoption of transformative technologies in the region.

Chief Strategy Officer @Techzoid (Pvt) Limited

Mar 2022- Mar 2023

As the Chief Strategy Officer at Techzoid (Pvt) Limited, I play a pivotal role in shaping and executing the company's strategic vision. My responsibilities include defining long-term goals, identifying growth opportunities, and overseeing the alignment of our business strategies with market trends. I am also responsible for fostering innovation and ensuring the company's overall success in the dynamic tech landscape.

Technology Support Engineer at HBL (Pvt) Limited

2019-2021

– As a Technology Support Engineer at Habib Bank Limited from June 2019 to November 2021, I managed diverse technical responsibilities encompassing network maintenance, system configuration, and comprehensive IT tasks. This included proficiently handling software installations and updates, contributing to seamless operational efficiency.

PROJECTS

Generative AI and RAG based project

[GitHub](#)

The following major tasks are performed in these projects.

- Fine-tuning of randomized clinical trials using various language models including Gemini, OpenAI, LLAMA, and Mistral.
- Implementing RAG (Retrieval-Augmented Generation) system for advanced medical chat functionalities focused on diseases and pharmaceuticals utilizing specialized medical language models.
- Developing multi-modal RAG systems integrating diverse data types and multi-modal language models.
- Utilizing generative AI techniques to innovate in healthcare, enhancing diagnostic and treatment processes

AI-Powered Document Extraction and Semantic Search (MVP)

[GitHub](#)

Major Tools and Technologies used in this project.

Developed an AI-driven document intelligence demo for Streamline Verify, featuring OCR-based data extraction from scanned PDFs, GenAI-powered summarization (Llama 3.23B), and semantic provider type matching using embeddings. The system runs fully offline with free/open-source tools, achieving fast, structured extraction and clustering. Technologies: Python, Tesseract, Llama 3.23B, MedEmbed, SentenceTransformers, FAISS, FastAPI, SQLite, React, matplotlib, plotly, regex, spaCy.

Frid.ai (Personal Information Butler)

The following major tasks are performed in this project.

Automatic daily life tasks like emails, and tweets.

Developing a Generative AI model with the help of Open AI to summarize the emails and tweets daily and convert all the emails and tweets into audio in a click application.

FACIAL Recognition USING CNN — AI LOUNGE— 2023

The following major tasks are performed in this project.

Develop an innovative Facial Expression Recognition system utilizing Convolutional Neural Networks (CNNs) at AI Lounge. Enhance emotion detection accuracy through deep learning, empowering applications in human-computer interaction and sentiment analysis.

NEXGEN Security AI Surveillance System

The following major tasks are performed in this project.

Develop an innovative facial expression recognition system utilizing object detection and object classification using YOLO and convolutional neural networks (CNNs). Enhance the surveillance system by detecting objects and generating alarms for breaches.

Prospect.AI

The following major tasks are performed in this project.

Core Offer: Automate real DM conversations that convert into bookings—no paid ads, just personalized outreach.

Target Clients: Service-based small businesses seeking scalable, low lift lead acquisition.

End Goal: Manage 10–100+ clients by automating lead sourcing, AI messaging, calendar booking, and reporting with minimal manual effort.

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Email Automations

The following major tasks are performed in this project.

Core Offer: Automate real time Reply of email conversations by using Large Language Models (LLM).

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EDUCATION

2024–2026	MS in Data Science from National University of Computer and Emerging Sciences (FAST NUCES Islamabad)	
2015–2019	BS(Hons)(Computer Science) at Karakorum International University	(GPA: 3.01/4.0)
2014	ICS from Army Public School and College, Gilgit	(B)
2012	Class 10th, from Federal Board of intermediate and Secondary School, Islamabad	(A+)

CERTIFICATIONS

1. Artificial Intelligence Developer

The One-Year Diploma Program in Artificial Intelligence and Computing was initiated by the Pakistani government's Presidential Initiative.

2. Natural Language Processing Specialization

[Certificate](#)

NLP Specialization Certificates include the following certificates.

1. Natural Language Processing with Classification and Vector Spaces.
2. Natural Language Processing with Probabilistic Models.
3. Natural Language Processing with Sequence Models.
4. Natural Language Processing with Attention Models.

3. Generative AI with Large Language Models

[Certificate](#)

Generative AI use cases, project lifecycle, and model pre-training.

Fine-tuning and evaluating large language models.

Reinforcement learning and LLM-powered applications.

4. Data Scientist with Python

[Certificate](#)

Data Camp Certificate mainly focuses on Python and Data Science concepts.

5. Data Analytics and Artificial Intelligence

3-month Diploma in AI and Data Analytics covered the following tools and skills. Power Bi, SQL, Python, Machine Learning.

 [LinkedIn Certifications](#)

SKILLS

Python

Machine Learning

Deep Learning

Data Visualization

Generative AI

Retrieval Augmented Generation (RAG)

Fine-tuning LLM Models

Agents AI

AI Automation